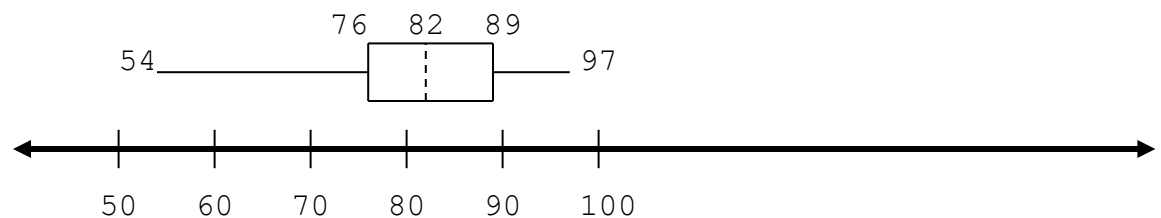


Math 103

Class 9

1. Box and Whisker Plots (boxplots)

Data items:	97	96	91	90	88	87	85	83	High	97
	81	80	79	77	75	68	60	54	Low	54
									Q1 (LQ)	76
									Q2 (MD)	82
									Q3 (UQ)	89
									1.5(IQR)	19.5
									Outliers	54



The difference between the upper and lower quartiles is called the interquartile range (IQR). An outlier is any value that lies more than $1.5(\text{IQR})$ units above or below the "box".

$$89 - 76 = 13$$

$$1.5(13) = 19.5$$

In this example $76 - 19.5 = 56.5$. Therefore, 54 is an "outlier" as it is less than 56.5.

$$89 + 19.5 = 108.5$$

Since there are no data items above 108.5, there are no "outliers" on the right end.

2. With an "even number" of data items, say $2n$, the lower quartile is the median of the n smallest data items. The upper quartile is the median of the n largest data items.
3. With an "odd number" of data items, say $2n+1$, the lower quartile is the median of the n smallest data items, and the upper quartile is the median of the n largest scores.

Example:

Data items: 49 53 65 68 70 75 78 80
81
83 85 88 90 94 95 95 96

High= 96

Low = 49

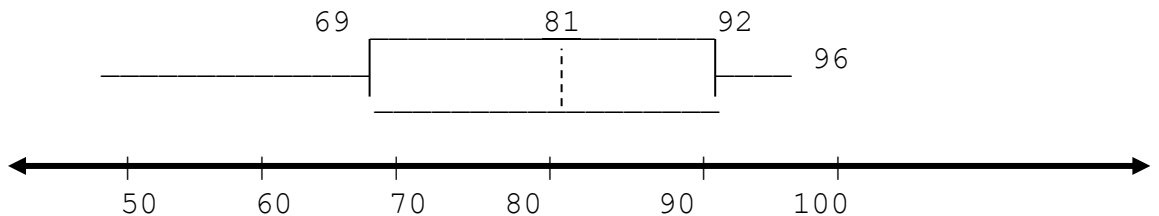
$$Q1 \text{ (LQ)} = \frac{68+70}{2} = 69$$

$$Q2 \text{ (MD)} = 81$$

$$Q3 \text{ (UQ)} = \frac{90+94}{2} = 92$$

$$1.5 \text{ (IQR)} = 92 - 69 = 23$$

$$\text{Outliers} = 1.5 (23) = 34.5$$



4. **Percentile** -- For whole numbers P (where $1 \leq P \leq 99$), the P^{th} **Percentile** of a distribution is a value such that P% of the data fall at or below it and $(100 - P)\%$ of the data fall at or above it. (See pp 134-135 in Brase)

Percentile Formula:

The percentile corresponding to a given value X is computed by using the following formula:

$$\text{Percentile} = \frac{(\text{Number of values below } X) + 0.5}{(\text{Total number of values})} \cdot (100\%)$$

Example: The scores of 10 students are given by $X = \{18, 15, 12, 6, 8, 2, 3, 5, 20, 10\}$.

- a) Find the percentile rank of a score of 12?

Solution: First, arrange the data in order from lowest to highest: 2, 3, 5, 6, 8, 10, 12, 15, 18, 20.

$P = \frac{6 + 0.5}{10} \cdot (100\%) = 65^{\text{th}}$ percentile. Thus, a student whose score was 12 did better than 65% of the class.

b) Find the percentile rank for a score of 6?

Solution: There are three values below 6. Now $P = \frac{3+0.5}{10}(100\%) = 35 \text{ percentile}$. A student who scored 6 did better than 35% of the class.

c) Find the value corresponding to the 25th percentile?

Solution: Here $C = \frac{n.p}{100}$, where, $n = \text{Total number of value}$, and $p = \text{percentile}$.

Thus $C = \frac{(10)(25)}{100} = 2.5 \approx 3$. So start at lowest value and count over to the third value, which is 5. Hence the value 5 corresponds to the 25th percentile.

d) Find the value that corresponds to the 60th percentile.

Solution: Now $C = \frac{(10)(60)}{100} = 6$. If C is a whole number, use the value halfway between the C and $C+1$ values when counting up from the lowest value, the 6th and 7th value. The value halfway between 10 and 12 is 11. Hence, 11 correspond to the 60th percentile. Anyone scoring 11 would have done better than 60% of the class.

5. Homework:

Make a box and whisker plot for these data items:
Identify any "outliers".

25	22	7	24	26	31	18	14	17	20
31	42	6	25	22	3	29	32	15	72

High	_____
Low	_____
Q1	_____
Q2	_____
Q3	_____
1.5 (IQR)	_____
Outliers	_____